Abigail Ollson

PhD student
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GitHub: **AbiOllson**

Education

- September 2023 – present: **PhD in Mathematics**, Keele University, UK. Supervisors: Dr. Christian Bean and Dr. Paul C. Bell.

  Thesis: *'Enumerating pattern-avoiding combinatorial objects defined as words on* $\mathbb{N}$*'* on methods for the automatic enumeration of sets of combinatorial objects defined as words on $\mathbb{N}$ such as set partitions, inversion sequences, and restricted growth functions avoiding Cayley permutations.

- September 2019 – June 2023: **MMath in Mathematics with competency in Japanese (First Class Honours)**, Keele University, UK. Supervisor: Dr Paul Truman. Dissertation: *'Error-Correcting Codes'*

- September 2017 – June 2019: **A-Levels**, Farmor's School, Fairford, UK. Subjects: Mathematics (A*), Further Mathematics (A*), Physics (A).

Publications

Journal articles:

- Bean, C., and Bell, P.C., Ollson, A. (2026). *The insertion encoding of restricted growth functions*, Annals of Combinatorics,10.1007/s00026-026-00832-y
- Celano, K. Ollson, A., Velankar, N. and Yan, J. (2025). *An Erdős-Szekeres type result for words with repeats*, Discrete Mathematics & Theoretical Computer Science, 10.46298/dmtcs.16801
- Bean, C., Bell, P., and Ollson, A. (2026). *The Insertion Encoding of Cayley Permutations*. The Electronic Journal of Combinatorics, 10.37236/14191

In preparation:

- Bean, C., Ollson, A., and Pantone, J. *Automatic discovery of generating trees for pattern-avoiding inversion sequences*
- Bean, C., and Ollson, A. *Combinatorial Exploration of Cayley Permutations*
- Bean, C., Ironmonger, V., and Ollson, A. *WQO for Cayley permutations under the consecutive order*

Other roles and achievements

- Advanced knowledge of Python, see my GitHub **AbiOllson**
- Advance HE Associate Fellow (AFHEA) (2024)
- Piscopia local committee member at Keele University (2025–current)
- PGR representative for Mathematics department at Keele University (2024–2026)

- Keele SU Volunteering Platinum Award (2023)
- President (2021–2022) and Vice-President (2020–2021) of Keele University Christian Union
- Data Analyst Intern at DCO Systems (2018–2019 and 2021).

  DCO Systems provides self-sustaining sensors to monitor and improve the efficiency of machines. My main responsibilities included interpreting data from large data sets, developing statistical models in R and Python and conducting experiments to gather data.

Teaching

Laboratory demonstrator at Keele University (2024–2026) for the following modules.

- Introduction to Algorithms (CSC-10064), *Spring 2026*
- Linear Algebra (MAT-10073), *Spring 2026*
- Guided Studies – Combinatorics (MAT-30053), *Autumn 2025*
- Mathematical Foundations of Computing (MAT-10067), *Autumn 2025*
- Introduction to Algorithms (CSC-10064), *Spring 2025*
- Algebra II (MAT-10040), *Spring 2025*
- Discrete Mathematics (MAT-10065), *Autumn 2024*
- Introduction to Algorithms (CSC-10064), *Spring 2024*

Private tutor for GCSE and A-Level Mathematics (2020–2026).

Conferences

Co-organiser of the Permutation Patterns conference at University of St Andrews (2025) and the PGR Symposium at Keele University (2026). Attended the following:

- Permutation Patterns, *California State University, Los Angeles, USA*, July 2026. **(talk)**
- Keele PGR Symposium, *Keele University, UK*, May 2026. **(co-organiser)**
- Postgraduate Combinatorics Conference, *University of Warwick, UK*, April 2026. **(talk)**
- Sequences, *City, University of London, UK*, Feburary 2026. **(talk)**
- PiForum, *Queen Mary University, UK*, September 2025. **(talk)**
- Permutation Patterns, *University of St Andrews, UK*, July 2025. **(talk, co-organiser)**
- Postgraduate Combinatorics Conference, *University of Glasgow, UK*, April 2025. **(talk)**
- Scottish Combinatorics Meeting, *University of Glasgow, UK*, April 2025. **(poster)**
- BOW Combinatorics Conference, *Open University, UK*, December 2024.
- PiForum, *University of Leicester, UK*, September 2024. **(talk)**
- International Conference on Enumerative Combinatorics and Applications (online), *University of Haifa, Israel*, August 2024.

- British Combinatorics Conference, *Queen Mary University, UK*, July 2024. **(talk)**
- Scottish Combinatorics Meeting, *University of St Andrews, UK*, May 2024. **(talk)**
- Postgraduate Combinatorics Conference, *University College London, UK and London School of Economics*, April 2024. **(talk)**